

trgn515_assign5_sub_github

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Task 1: Basecalling with modified bases

1. What is the output of the basecalling when using modified bases? (5pts)

- The output is a BAM file, not a FASTQ file. Because `--modified-bases-models` is included, Dorado needs to store the methylation information along with the sequencing data, and BAM format supports this while FASTQ does not.

2. Why do you think they have chosen this file type? (5pts)

- This is because BAM format supports auxiliary tags, which allows it to store additional information such as methylation data alongside the sequence and alignment information.
- FASTQ can only store sequence and quality scores.

3. What are the tags that represent methylation, and what do each of them mean? (5pts)

- There are two tags represent methylation in the BAM file.
- MM (Base Modification): Encodes the position and type of base modifications along the read, specifying which bases are modified and what type of modification they carry, such as 5mC or 5hmC.
- ML (Modification Likelihood): Stores the probability of each modification call as values between 0 and 255, representing the confidence level of the modification at each position.

4. In our Dorado command, we passed the argument `-r sup,5mCG_5hmCG`. What does `sup` and `5mCG_5hmCG` mean? (10pts)

- `sup` stands for Super Accuracy model, which is the most accurate basecalling model in Dorado. It gives better results than the fast or high accuracy models, but needs more computing power to run.
- `5mCG_5hmCG` is the model used in this assignment to detect two types of DNA modifications at the same time. 5mCG is 5-methylcytosine at CpG sites, which is the most common type of DNA methylation and is associated with gene silencing. 5hmCG is 5-hydroxymethylcytosine at CpG sites, which is an oxidized form of 5mC and acts as an intermediate in the DNA demethylation process.

5. Why have we chosen `5mCG_5hmCG` for a human dataset? (10pts)

- We chose `5mCG_5hmCG` for a human dataset because DNA methylation in humans predominantly occurs at CpG sites. 5mC is the primary epigenetic modification associated with gene regulation,

particularly gene silencing. 5hmC, an oxidized form of 5mC generated by TET enzymes, is involved in active DNA demethylation and also has independent regulatory roles. These two modifications are biologically relevant and well-characterized in human genomes, making 5mCG_5hmCG an appropriate model for this dataset.

Task 2: Mapping modified reads.

1. What does the -T flag in samtools fastq do and why is it important? (5pts)

- The -T flag in samtools fastq specifies which auxiliary tags from the BAM file should be retained when converting to FASTQ. Since FASTQ format does not natively support auxiliary tags, these tags (e.g., MM and ML) are encoded into the read header.
- This is important because methylation information is stored in these tags, and without the -T flag, this information would be lost during conversion. As a result, all modification-related data from basecalling would be discarded.

2. What does the -y (lower case!) flag in minimap2 do and why is it important? (5pts)

- The -y flag tells minimap2 to read tags embedded in FASTQ comments/header and retain them in the SAM/BAM output after alignment. It works together with the -T flag. -T keeps the methylation tags when going from BAM to FASTQ, and -y keeps them again when going from FASTQ back to BAM after alignment. Without -y, the methylation tags would be lost during mapping even though they were saved in the FASTQ. So both flags are needed together to make sure the methylation information is preserved throughout the whole conversion and alignment process.

Task 3: Make a pileup of the methylation calls

1. What are the columns of the resulting file? Visit <https://github.com/nanoporetech/modkit> for more information (5pts)

- 18 columns in total
- 1. chrom 2. start position, 3. end position 4. modified base code, 5. score, 6. strand
- 7. start position: included for compatibility, 8. end position: included for compatibility, 9. color
- 10. Nvalid_cov: total number of valid coverage reads, 11. fraction modified, 12. Nmod
- 13. Ncanonical, 14. Nother_mod: number of other modification calls, 15. Ndelete, 16. Nfail: number of failed calls, 17. Ndiff: number of different calls, 18. Nncall: number of no-call positions

2. Can you think on some filters that may be important given the provided columns? (5pts)

- I think some important filters would be Coverage (column 10, Nvalid_cov) to filter out positions with very low coverage, such as only 1 or 2 reads, because the methylation estimate would not be reliable.
- I would also filter out positions with a high Number of failed calls (column 16, Nfail) because this shows the position has very poor sequencing quality.
- I would filter based on Number of no-calls (column 18, Nncall) to remove positions where many reads could not be assigned a methylation status.

3. Open the resulting BED file and look at the first few rows. How many rows are generated for each single genomic position, and what specific column differentiates them? Based on our ModKit command, why is the file structured this way? How many rows do you expect per genomic position? (5pts)

- I could see that each genomic position generates 2 rows.
- For example, position chr1 10788 10789 appears twice. The column that differentiates them is column 4 (modified base code), where one row has h for 5hmC and the other has m for 5mC.
- This is because in ModKit command I used `-modified-bases 5mC 5hmC`, which tells ModKit to report each modification type separately for every CpG site. Therefore, I expect 2 rows per genomic position.

4. Look at column 11 (fraction modified). If a genomic region is fully methylated on both alleles, you would expect a value near 100%. What biological scenario would result in a fraction of ~50%? (5pts)

- A fraction of ~50% would most likely result from allele-specific methylation. One allele is fully methylated while the other allele is completely unmethylated. Since sequencing captures reads from both alleles equally, the resulting fraction modified would be approximately 50%. For an example, genomic imprinting, where one allele inherited from the mother is methylated and the other allele inherited from the father is not, or vice versa.

5. What column would you use to identify different methylation patterns in each allele? How would a heterozygous methylation pattern look like? (5pts)

- To identify different methylation patterns, I would use column 11 (fraction modified) together with column 10 (Nvalid_cov) and column 12 (Nmod). These columns describe the proportion of methylated reads at a given position, although they do not directly distinguish between alleles without phasing information.
- A heterozygous methylation pattern would appear as a fraction modified value of around 50%, meaning roughly half of the reads show methylation and the other half do not.
- For example, if coverage is 20 reads and Nmod is 10, this would indicate that approximately half of the reads are methylated and half are unmethylated, suggesting a mixed methylation pattern. This is different from a homozygous methylated region, which would show a fraction close to 100%, or a homozygous unmethylated region, which would show a fraction close to 0%.

6. Why do we run bgzip and tabix on the resulting BED file before moving to the next step? What do these tools do? (5pts)

- We run bgzip and tabix on the BED file because the next step, modkit dmr, requires the input files to be compressed and indexed for fast random access.
- bgzip compresses the BED file into a .bed.gz format. bgzip divides the file into small compressed blocks. This allows tools to jump directly to any specific position in the file without having to decompress and read the entire file from the beginning.
- tabix creates an index file (.tbi) that allows tools to quickly jump to any genomic region without reading the entire file.

Task 4: Interpreting the DMR Output

1. Which are the two genes with the highest differences? Which gene shows hypermethylation (increased methylation) and which shows hypomethylation (decreased methylation) in the tumor sample compared to the normal sample? Which columns did you use to determine this? (10pts)

- The two genes with the highest differences are KRT19 (score = 8244) and TPM2 (score = 2821), based on column 5 (statistical score).
- Looking at columns 11 and 12 (fraction modified in normal and tumor samples), as well as column 13 (effect_size), we can determine the direction of methylation changes.
- KRT19 shows hypomethylation in the tumor sample, with methylation decreasing from 0.751 in the normal sample to 0.326 in the tumor sample (effect_size = -0.425).
- In contrast, TPM2 shows hypermethylation in the tumor sample, with methylation increasing from 0.560 in the normal sample to 0.744 in the tumor sample (effect_size = +0.183).

2. Which of the previous two genes is most likely a tumor suppressor gene and which one is most likely an oncogene? How is this related to their methylation pattern? (10pts)

- TPM2 is likely a tumor suppressor gene because it shows hypermethylation in the tumor sample. When a tumor suppressor gene's promoter becomes hypermethylated, the gene gets silenced, which removes its ability to control cell growth and promotes cancer development.
- KRT19 is likely an oncogene because it shows hypomethylation in the tumor sample. When an oncogene becomes hypomethylated, it gets activated, which can drive uncontrolled cell proliferation and cancer progression.

3. Compare the results for KRT19 and GPX1. Both have high statistical scores (significance), but how do their effect_size values differ? Which one should we prioritize when looking for drivers of disease? Which one is more likely to be a statistical artifact? (5pts)

- Both KRT19 and GPX1 have high statistical scores, but their effect sizes are very different.
- KRT19 has a large effect size of -0.425, showing hypomethylation in the tumor sample compared to the normal sample. GPX1 has a very small effect size of only 0.016, even though its statistical score is high.
- KRT19 should be prioritized when looking for drivers of disease because it has both high significance and a large effect size. GPX1 is more likely to represent a statistically significant but weak biological signal, as small effect sizes can become significant due to high sequencing coverage.

4. Why is it statistically more robust to calculate differential methylation across a region (like a CpG island or promoter) rather than looking at individual, isolated cytosines? (5pts)

- It is more robust to calculate differential methylation across a region rather than at individual CpG sites because single sites can be noisy due to sequencing errors or low coverage.
- When methylation is averaged across a larger region, like a CpG island or promoter, the random variation is reduced and the overall signal becomes more reliable.
- Also, analyzing regions instead of individual sites reduces the number of statistical tests, which lowers the chance of false positives.

5. In this analysis, the coordinates used for the DMR covered the entire gene body for each gene. Explain why measuring average methylation changes across an entire gene might not be the most appropriate method for identifying biologically meaningful changes, and propose a better approach. (5pts)

- Measuring methylation across the entire gene body is not the most appropriate approach because different regions of a gene can have different roles in gene regulation.
- In this assignment, we are interested in how methylation relates to cancer, and changes in promoter regions or CpG islands are usually more important for controlling gene expression. In contrast, methylation within the gene body does not always have the same effect and can sometimes be associated with active genes.
- Thus, if we average methylation across the whole gene, these local changes may be diluted or missed. A better approach would be to focus on specific regions such as promoters or CpG islands near the transcription start site, where methylation changes are more likely to have functional consequences.

Task 5: Visualize modified bases

1. What are the differences in methylation patterns you observe between normal and tumor samples? Are they consistent with the DMR analysis? (10pts)



- The methylation patterns show clear differences between the normal and tumor samples. In the KRT19 region, the normal sample displays a higher level of methylation, as indicated by a greater proportion of red signals. In contrast, the tumor sample shows a noticeable increase in blue signals, indicating reduced methylation levels (hypomethylation). This pattern is consistent with the DMR analysis, which identified KRT19 as significantly hypomethylated in the tumor sample. Therefore, the visualization supports the differential methylation results observed in the DMR analysis.



- In the TPM2 region, the tumor sample shows a higher level of methylation compared to the normal sample. This is indicated by an increase in red signals and a relative decrease in blue signals in the tumor sample.
- This is consistent with the DMR analysis, which showed that TPM2 methylation increased from 0.560 in the normal sample to 0.744 in the tumor sample (effect size = +0.183), indicating hypermethylation in the tumor. However, the difference is less visually obvious compared to KRT19, which is expected given that TPM2 has a smaller effect size.



- Looking at the IGV visualization of TP53, the methylation pattern between the normal and tumor samples appears very similar. Both samples show predominantly hypermethylated reads throughout the gene body, with only minor differences in the distribution of hypomethylated reads. This is because even though TP53 has a high statistical score, its effect size is very small (0.035). The fraction modified only changed slightly from 0.924 in the normal sample to 0.959 in the tumor sample, meaning both samples are already highly methylated. The small difference between them is too subtle to be visually

obvious in IGV. The high statistical score is likely driven by the very high sequencing coverage of this region rather than a large biological difference, which highlights why effect size is just as important as statistical significance when interpreting DMR results.

2. Do the methylation changes occur throughout the entirety of the gene or in a localized area? Discuss how these visual results affect our whole-gene DMR analysis and whether you would modify it in light of the IGV check. (10pts)

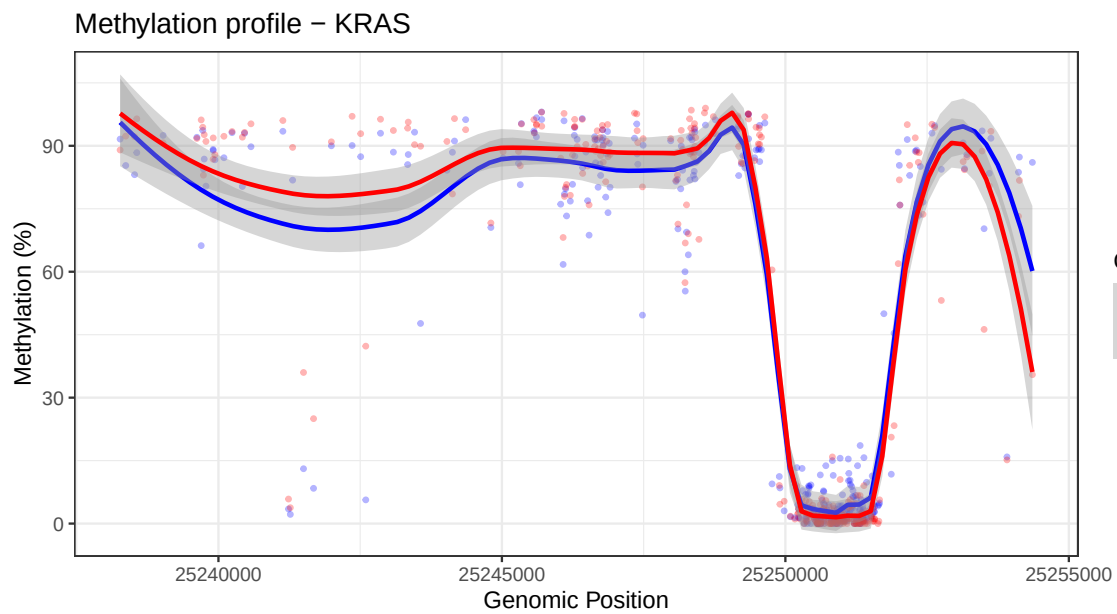
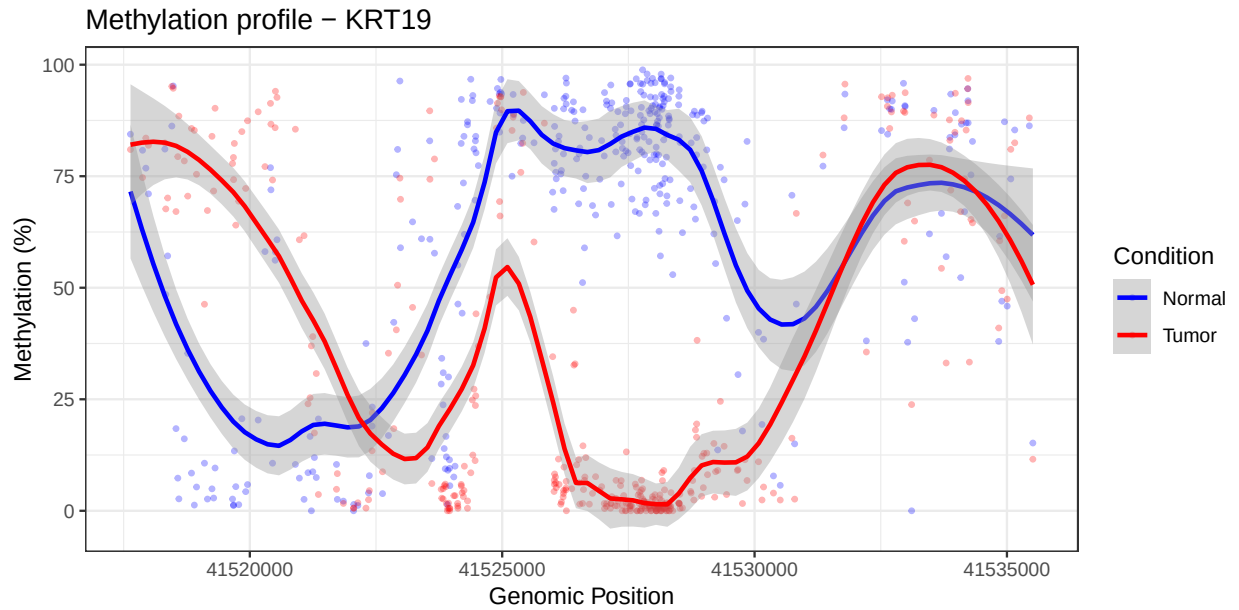
- KRT19: Looking at the IGV visualization, the methylation changes in KRT19 do not occur throughout the entirety of the gene. The main difference between normal and tumor is around the promoter region. The normal sample shows higher methylation, while the tumor sample shows a clear decrease in methylation. The rest of the gene does not show such strong differences.
- TPM2: for TPM2, the methylation changes appear to be more concentrated in a specific region of the gene rather than uniformly distributed across the entire gene body.
- How this affects our whole-gene DMR analysis: These results suggest that using the whole gene for DMR analysis might not be the best approach. When we average methylation across the entire gene, we are diluting the signal from the regions that are actually changing. The localized changes at the promoter region are more biologically meaningful but get averaged out with the less informative gene body regions.
- Would I modify the analysis? Yes, I would modify the analysis to focus specifically on the promoter regions and CpG islands near the transcription start site rather than the entire gene body. This would give a more accurate and biologically meaningful picture of the differential methylation changes associated with cancer.

3. Navigate to the TP53 gene and answer the following questions: This gene had a high statistical score (794). Do the methylation tracks between the tumor and normal sample look drastically different to the naked eye? Why? (5pts)

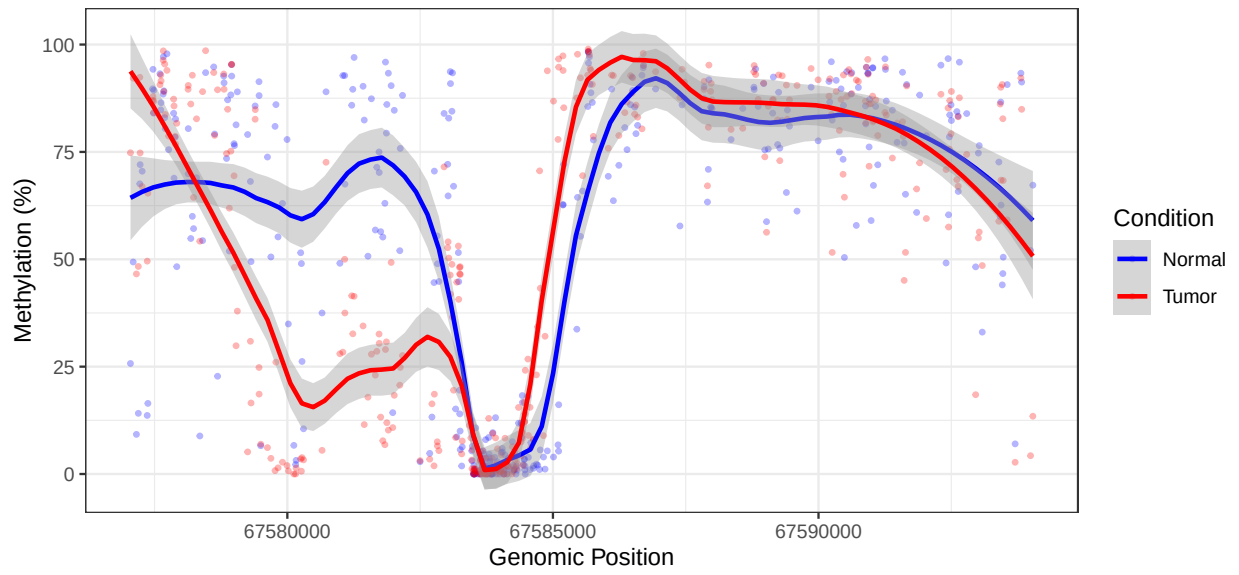
- Based on my IGV visualization of TP53, the methylation tracks between the normal and tumor samples do not look drastically different to the naked eye. Both samples show predominantly hypermethylated reads throughout the gene body, with only minor differences visible.
- This is because even though TP53 has a high statistical score, its effect size is very small (0.035). The fraction modified only changed slightly from 0.924 in the normal sample to 0.959 in the tumor sample, meaning both samples are already highly methylated to begin with. The difference between them is too small to be visually obvious in IGV. This is a good example of why statistical significance alone is not enough — a high score can be driven by very high sequencing coverage rather than a true large biological difference, which is why effect size is just as important when interpreting DMR results.

Task 6: Loess methylation plot.

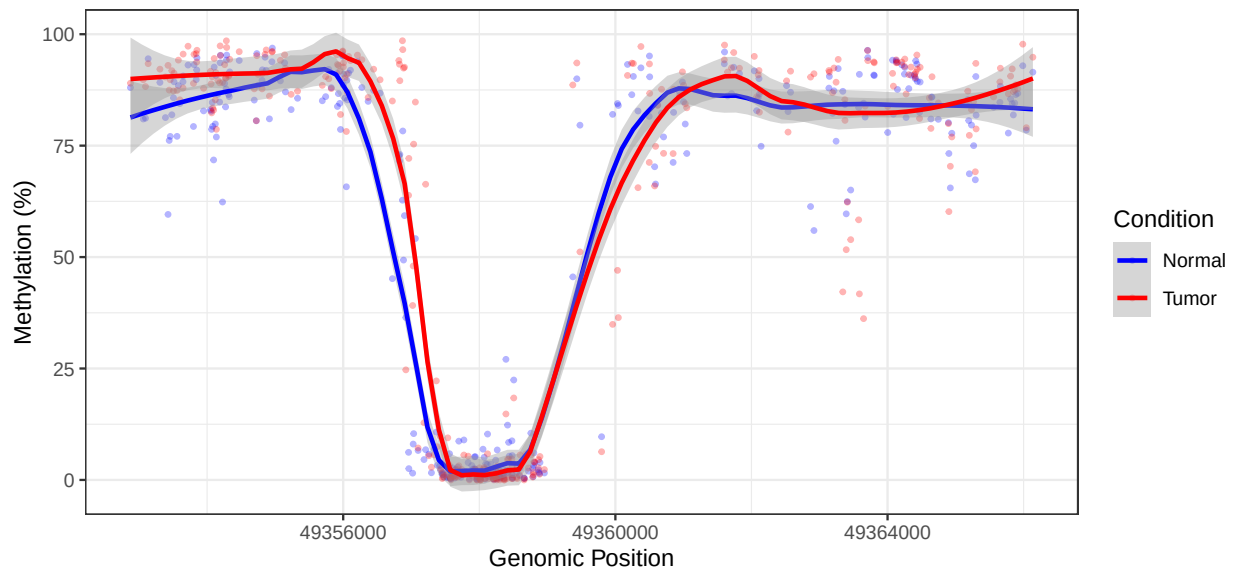
1. Include the generated plots for all 8 genes (15pts)

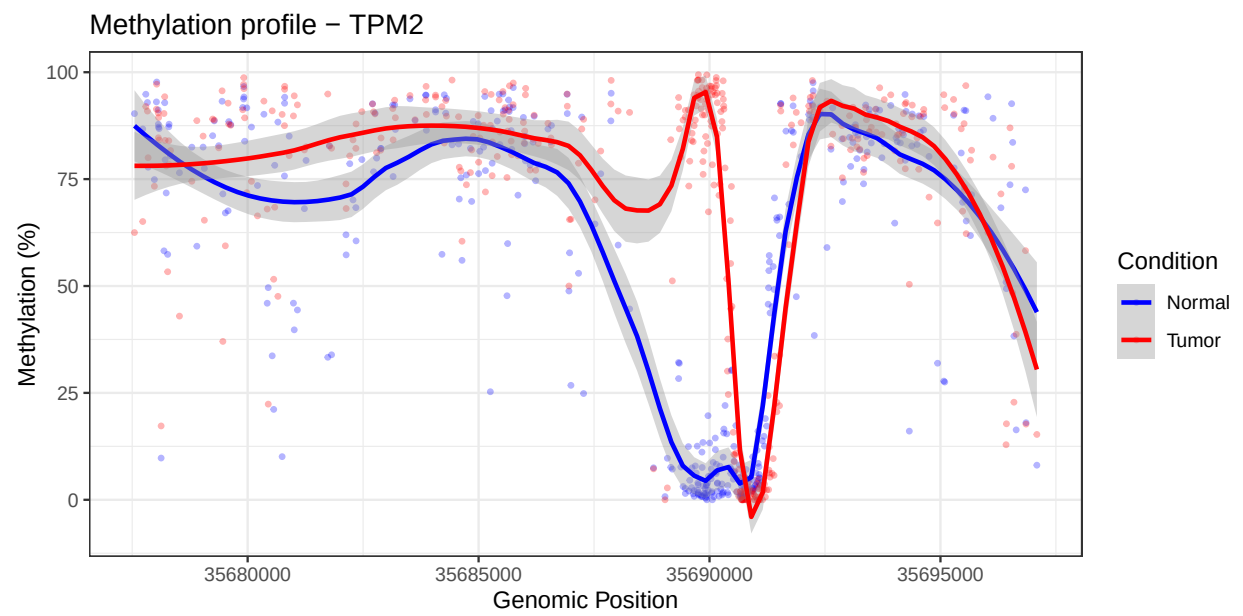
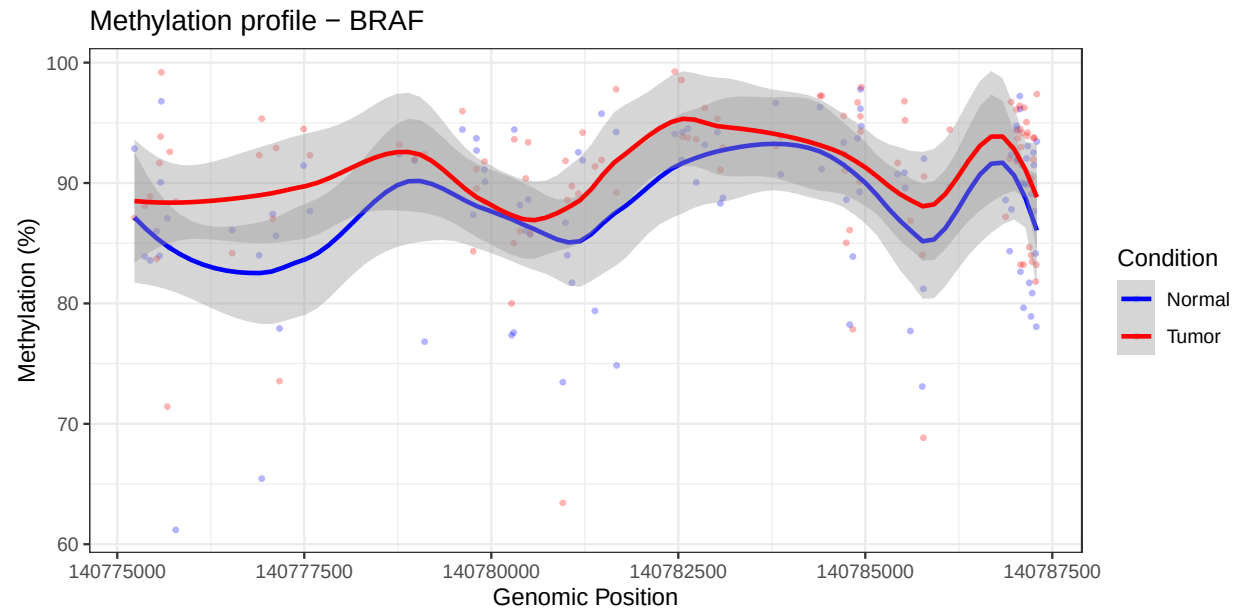


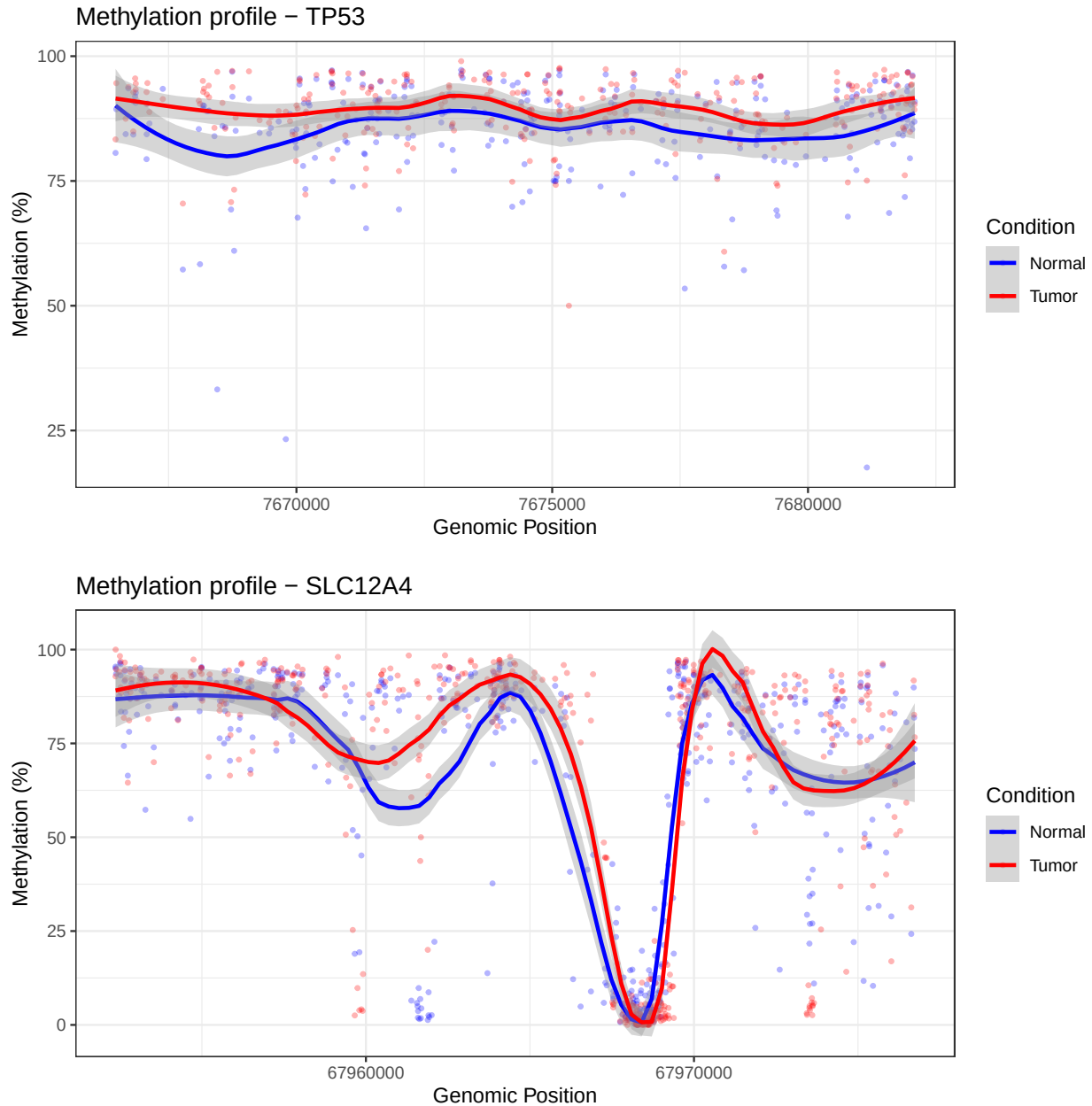
Methylation profile – GSTP1



Methylation profile – GPX1







2. Look at the scatterplot dots underlying the smoothed curves. Do all neighboring sites share the exact same methylation percentage, or is there high variance from one nucleotide to the next? What does this tell you about the biological “noise” of DNA methylation? (10pts)

- Looking at the scatterplot dots across all methylation profiles (e.g., TP53, KRT19, TPM2, KRAS, GSTP1, GPX1, BRAF, and SLC12A4), neighboring CpG sites do not share the exact same methylation percentage. Instead, there is clear variation from one nucleotide to the next, with individual points scattered above and below the smoothed curves.
- In TP53 and BRAF, most sites are highly methylated, but there is still some spread between nearby CpG sites. In contrast, genes like KRT19, TPM2, and GSTP1 show much larger variation, especially in regions where methylation levels change rapidly, with some CpG sites showing very low methylation

while nearby sites are much higher.

- This suggests that DNA methylation is quite variable at the single-CpG level, reflecting a high level of biological noise. This could be due to real biological differences (like variation between cells), as well as technical factors such as sequencing noise or uneven coverage.
- The loess-smoothed curves help smooth out this variation and make it easier to see the overall trend across the region, which is more useful than looking at individual CpG sites.

3. Compare the plot for a gene with a big effect size (like KRT19 or TPM2) to a gene with an effect size near zero (like KRAS). How does the distance between the Tumor and Normal LOESS curves reflect the effect size calculated in Task 4? (10pts)

- Comparing the LOESS plots, all genes show some separation between the Normal and Tumor curves, but the degree of separation varies and generally reflects the effect size calculated in Task 4.
- For KRT19 (effect size = -0.425), the Tumor curve is generally lower than the Normal curve in many regions, indicating hypomethylation in the tumor sample. The separation is large in several parts of the gene, especially in the central region, although it is not completely uniform across the entire gene.
- For TPM2 (effect size = +0.183), the Tumor curve is often higher than the Normal curve in parts of the region, indicating hypermethylation in the tumor sample, with a moderate level of separation between the curves.
- For KRAS (effect size = 0.002), there is still some visible difference between the curves, but the separation is generally much smaller and less consistent. The curves often follow a similar trend and remain relatively close to each other, indicating minimal overall methylation change.
- Overall, the distance between the Tumor and Normal LOESS curves reflects the effect size in a relative way. This larger and more consistent separation corresponds to larger effect sizes, while smaller or more localized differences correspond to effect sizes close to zero.

4. Compare the profiles of TPM2 and KRT19. Which one shows broad methylation changes and which one shows localized ones? (5pts)

- Comparing the methylation profiles of KRT19 and TPM2, KRT19 shows broader methylation changes, while TPM2 shows more localized changes. In KRT19, the Tumor curve is generally lower than the Normal curve across multiple regions of the gene, indicating widespread hypomethylation. In contrast, TPM2 shows more localized changes, where the main difference between the two curves is concentrated around a central dip region, while the rest of the gene shows less separation.